



Sem:IV

Subject Code: BP402T

Date:27/08/2022

Duration: 3.45 Hr.

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = $20 \times 1 = 20$ (All the questions are compulsory)

i) Gaseous and volatile drugs and their metabolites are excreted through

- a) Skin b) Lungs

- c) Billiary route

ii) Major site of drug metabolism is

- a) Lung
b) Liver
c) Skin
d) Gastrointestinal tract

iii) Conversion of alcohols to aldehydes is _____ metabolic reaction.

- a) Oxidation b) Reduction
c) Hydrolysis d) Conjugation.

iv) Enzyme used for glucuronic acid conjugation is _____.

- a) Glutathione S-transferase b) UDP-Glucuronyl transferase
c) SAM d) PAPS

v) Henderson-Hasselbalch equation is used to determine pH of drug.

- a) Partition Coefficient b) Solubility
c) % ionization d) pH.

vi) % ionisation of drug depends upon _____

- a) pKa of the drug b) pH of the body fluid
c) Partition coefficient of drug d) Both a. and b.

vii) β_2 receptors are found in_____.



- a) Heart
c) Adipose tissue
viii) Muscarinic receptors are _____ receptor.
a) Nuclear
c) Enzyme
ix) Which of the following drug is a direct acting sympathomimetic drug.
a) Phenylephrine
c) Propylhexedrine
x) Which of the drug is alpha adrenergic blocker?
a) Propranolol
c) Atenolol
xi) Acetylcholine on hydrolysis gives....
a) Serine and acetic acid
c) Acetic acid and serine
xii) Chlordiazepoxide, a psychotherapeutic agent belongs to class of _____.
a) Carbamates
c) Benzodiazepine
xiii) Acetylcholine is biosynthesized from _____.
a) L-Cysteine
c) L-Serine
xiv) Hypnotics are _____.
a) Strong depressant of CNS
c) Mild depressant of CNS
xv) Chemical name 5-ethyl, 5-phenyl barbituric acid belong to
a) Hexobarbital
c) Pentobarbital
xvi) Which one of the following receptor is block by Neuroleptic agent?
a) Cholinergic
c) GABA
xvii) Which of the Anticholinergic agent useful as a spasmolytic drug?
a) Pyridostigmine
- b) Bronchi
d) Skeletal muscles.
b) G-protein Coupled
d) Ligand gated Ion Chanel.
b) Pseudoephedrine
d) Ephedrine
b) Metibranolol
d) Tolazoline
b) Choline and carbamic acid
d) Choline and acetic acid
b) Propanediol
d) Phenothiazine.
b) L-Codeine
d) L-Cholic acid
b) Strong stimulant of CNS
d) Mild stimulant of CNS
b) Phenobarbital
d) Secobarbital
b) Adrenergic
d) Dopaminergic
b) Dicyclomine

c) Tropicamide

d) Glycopyrrolate

xviii) Phenobarbital is orally administered in the treatment of:

a) Grand mal epilepsy

b) Petit mal epilepsy

c) Jackson epilepsy

d) Psychomotor epilepsy.

xix) Which of the following barbiturate derivative acts as anticonvulsant agent?

a) Hexobarbital

b) Phenobarbital

c) Pentobarbital

d) Secobarbital

xx) Which of the following drugs is acts as Cholinergic Blocking agent?

a) Physostigmine

b) Neostigmine

c) Pyridostigmine

d) Dicyclomine

Q. 2. Long Answers = $2 \times 10 = 20$ (Answer 2 out of 3)

i) Discuss the various physicochemical properties affecting the biological activity of drugs.

ii) Classify sedatives and hypnotics. Add a note on SAR and MOA of Barbiturates.

iii) Discuss the SAR of Parasympathomimetic agents and add a note on cholinergic receptors.

Q. 3. Short Answers = $7 \times 5 = 35$ (Answer 7 out of 9)

i) Discuss SAR of sympathomimetic drugs. Add a note on direct acting sympathomimetic drugs.

ii) Describe SAR of Morphine analogues. Add a note on Narcotic antagonists.

iii) Describe the factors that affect drug metabolism.

iv) Define Epilepsy and describe general mechanism of action of anticonvulsant drugs.

v) Classify anticonvulsant drugs and add a note on hydantoin derivatives.

vi) Classify antipsychotic drugs. Add a note on SAR and MOA of phenothiazine derivatives.

vii) Write a note on Biosynthesis and catabolism of catecholamine.

viii) Outline the synthesis of Phenytoin and Barbitol.

ix) Write a note on adrenergic receptor. Classify sympathomimetic drugs with examples

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